

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 20%

QUESTIONS: 2 Total Marks:20 Annexure A

Comparative Analysis

Code of Conduct:

I K. Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

*action. **Signature of the student:** K. Pujitha*

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies	
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive	
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided	
Synthesis & Insight Generation	20%	Generates meaningful insights and	Some insights derived from comparison	Limited or unclear insights	No insights generated	

		conclusions from comparisons				
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand	

Annexure A

Comparative Analysis

1. CNNs vs. Vision Transformers (ViTs) for Object Detection Scenario:

A self-driving car company needs to detect rare urban obstacles (traffic cones, animals) in real-time using limited labeled datasets.

Questions:

- Compare CNNs and Vision Transformers in terms of:
 - Feature extraction capabilities for local vs. global context
 - Performance with small or sparse objects
 - Computational complexity and inference speed for embedded systems
- Discuss which method is better suited for real-time deployment and how transfer learning or data augmentation could impact performance.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario: An animation studio is aligning 3D human face meshes for morphing applications. Meshes vary in topology and vertex density.

Questions:

- Compare PCA-based shape priors and graph-based learning in terms of:
 - Ability to handle non-rigid deformations
 - Robustness to varying mesh resolutions and noise
 - Ease of implementation and computational cost
- Analyze which approach would provide more accurate correspondences for realistic morphing and why.

1.Comparative Analysis

CNNs vs. Vision Transformers (ViTs) for Object Detection

Scenario

A self-driving car company needs to detect rare urban obstacles, such as traffic cones and animals, in real time. The main challenges are the limited number of labeled images, small or distant obstacles, complex urban backgrounds, and the computational limitations of embedded systems.

1. Selection of Studies / Parameters

The comparison is based on the following important parameters:

1. Feature extraction capability
2. Local and global context understanding
3. Performance on small and sparse objects
4. Performance with limited training data
5. Computational complexity
6. Inference speed
7. Suitability for embedded systems
8. Transfer learning and data augmentation
9. Advantages and limitations

2. Comparison of CNNs and Vision Transformers

Parameter	CNNs	Vision Transformers (ViTs)
Basic principle	Uses convolution operations to extract image features	Uses self-attention to model relationships between image patches
Feature extraction	Excellent at extracting local features such as edges, textures, and shapes	Captures both local and long-range relationships between image regions
Local context	Very strong because convolution focuses on neighboring pixels	Can capture local context, but requires suitable architecture or attention mechanisms

Parameter	CNNs	Vision Transformers (ViTs)
Global context	Global information is learned gradually through deeper layers	Strong global-context understanding through self-attention
Small objects	Effective with multi-scale feature extraction and feature pyramids	Can capture relationships between small objects and their surrounding regions
Sparse objects	Good when suitable detection and multi-scale techniques are used	Strong ability to relate sparse objects to the overall scene
Limited dataset	Generally more suitable, especially with transfer learning	Usually benefits strongly from pretraining because large Transformers can require substantial data
Computational complexity	Generally lower and more efficient	Generally higher, particularly for high-resolution images
Inference speed	Usually faster on embedded hardware	Can be slower depending on model size and implementation
Memory requirement	Generally lower	Generally higher
Embedded deployment	Highly suitable for resource-constrained devices	Possible with lightweight/optimized models but more challenging
Main advantage	Efficient local feature extraction and fast inference	Strong global-context and long-range relationship modeling
Main limitation	May require deeper networks or multi-scale methods for strong global understanding	Higher computational and memory requirements

3. Feature Extraction: Local vs. Global Context

CNNs

CNNs use convolution filters to process neighboring pixels. Therefore, they are very effective at identifying local features such as:

- Edges
- Corners
- Textures
- Shapes
- Object boundaries

As the network becomes deeper, the receptive field increases and the model can learn larger-scale information.

For example, a CNN can identify the edges and shape of a traffic cone even when it occupies only a small region of the image.

Vision Transformers

ViTs divide an image into patches and use self-attention to learn relationships between different patches.

This allows the model to understand global context.

For example, the model can consider:

Small obstacle + road + vehicle + surrounding environment

rather than examining the obstacle independently.

Critical Comparison

CNNs have an advantage in efficient local feature extraction, while ViTs have an advantage in modeling long-range relationships and global scene context.

4. Performance on Small or Sparse Objects

Detecting small obstacles is difficult because they occupy only a small number of pixels.

CNN Approach

CNN-based object detectors can use:

- Multi-scale feature extraction
- Feature pyramids
- High-resolution feature maps

- Data augmentation

These techniques help detect small objects such as distant traffic cones and animals.

ViT Approach

ViTs can use attention mechanisms to understand relationships between an object and its surrounding scene. This can help when the object is small or visually similar to the background.

However, performance depends strongly on:

- Model architecture
- Pretraining
- Image resolution
- Amount and quality of training data

Comparison

For a small labeled dataset, a pretrained CNN-based detector is generally a practical choice. A pretrained ViT can also perform well, particularly when global context is important, but its computational requirements may be higher.

5. Computational Complexity and Inference Speed

CNNs

CNNs are computationally efficient because convolution operations can be highly optimized on GPUs and embedded AI hardware.

Advantages include:

- Lower memory requirements
- Faster inference
- Efficient hardware acceleration
- Suitable for real-time applications

Vision Transformers

ViTs use self-attention, which can become computationally expensive as image resolution and the number of image patches increase.

High-resolution images such as 1024×768 can therefore increase computational requirements.

However, lightweight Transformer architectures and efficient attention mechanisms can reduce this problem.

6. Suitability for Real-Time Deployment

For a self-driving car, the system must provide accurate predictions with very low delay.

CNNs are generally better suited when:

- The dataset is relatively small.
- Real-time inference is required.
- Embedded hardware has limited computing power.
- Low latency is important.
- Efficient object detection is required.

ViTs are preferable when:

- Sufficient computational resources are available.
- Strong global-context understanding is required.
- A suitable pretrained model is available.
- Accuracy is prioritized along with efficient optimization.

Recommended Choice

For the given scenario, a lightweight pretrained CNN-based object detector is the most practical choice for initial real-time deployment.

A Vision Transformer can be considered when the hardware is powerful enough and the model has been properly optimized.

7. Effect of Transfer Learning

Since only a limited number of labeled obstacle images are available, training a large model completely from scratch is not ideal.

Transfer learning can be used as follows:

Pretrained Model → Replace/Adapt Detection Head → Fine-tune on Rare Obstacles → Test

A model pretrained on a large image dataset already knows general visual features such as edges, textures, and shapes.

This can:

- Reduce training time
- Improve accuracy
- Reduce the amount of labeled data required
- Improve generalization

Transfer learning is particularly useful for rare-object detection.

8. Effect of Data Augmentation

Data augmentation can increase the diversity of the training dataset.

Useful techniques include:

- Rotation
- Horizontal flipping
- Scaling
- Cropping
- Brightness adjustment
- Contrast adjustment
- Noise addition
- Synthetic image generation

For rare obstacles, augmentation can create examples with different sizes, positions, lighting conditions, and backgrounds.

This reduces overfitting and improves the model's ability to recognize unseen situations.

9. Critical Evaluation

CNN Strengths

- Fast inference

- Efficient feature extraction
- Lower computational requirements
- Good performance with transfer learning
- Suitable for embedded systems
- Effective for small-object detection with multi-scale features

CNN Weaknesses

- Global relationships may require deeper architectures
- Performance can decrease when objects are highly occluded or surrounded by complex backgrounds
- May require specialized multi-scale techniques for very small objects

ViT Strengths

- Excellent global-context modeling
- Captures long-range relationships
- Can be effective in complex scenes
- Strong performance when pretrained and properly optimized

ViT Weaknesses

- Higher computational and memory requirements
- Can be challenging for resource-constrained embedded devices
- Large models may require substantial pretraining
- High-resolution images can increase computational cost

10. Synthesis and Insights

The comparison shows that there is no single method that is best for every situation.

For the given self-driving car scenario:

- CNNs are more practical for real-time embedded deployment.
- ViTs provide stronger global-context modeling but generally require more computational resources.
- Transfer learning is important because the labeled dataset contains only a limited number of rare obstacles.
- Data augmentation can increase the diversity of rare-object examples.

- A lightweight architecture should be selected to balance accuracy, speed, memory usage, and safety requirements.

A useful practical pipeline is:

Camera Image → Preprocessing → Data Augmentation → Pretrained Lightweight CNN → Object Detection → Bounding Box + Confidence Score → Vehicle Decision System

11. Performance Evaluation

The final system should be evaluated using:

Precision

$\text{Precision} = \frac{TP}{TP + FPP}$

Measures how many detected obstacles are actually correct.

Recall

$\text{Recall} = \frac{TP}{TP + FNP}$

Measures how many actual obstacles are successfully detected.

F1-Score

$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

IoU

$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}}$

Inference Speed

Measured using FPS (frames per second) or latency per frame.

For autonomous driving, high recall and low inference latency are especially important because missing a rare obstacle can have serious safety consequences.

2. PCA-Based Shape Priors vs. Graph-Based Learning for 3D Shape Correspondence

Scenario

An animation studio needs to align 3D human face meshes for realistic facial morphing applications. The input face meshes may have different topologies, vertex densities, and shapes. Therefore, an accurate correspondence method is required to match equivalent facial regions between different meshes.

1. Comparison of PCA-Based Shape Priors and Graph-Based Learning

Parameter	PCA-Based Shape Priors	Graph-Based Learning
Basic Principle	Learns the mean shape and major variations from a set of training shapes using PCA.	Represents the 3D mesh as a graph and learns relationships between connected vertices.
Non-Rigid Deformations	Handles common and relatively small shape variations well, but may struggle with complex facial deformations outside the learned PCA space.	Better suited for complex non-rigid deformations because it learns local and global relationships between mesh vertices.
Varying Mesh Resolution	Works best when shapes are aligned and represented using compatible points or landmarks. Different vertex counts can require preprocessing or resampling.	Can naturally represent meshes with different numbers of vertices because each mesh can be represented as an individual graph.
Different Topology	Sensitive to differences in topology and point correspondence.	More flexible for different mesh topologies because graph structure represents connectivity within each mesh.

Parameter	PCA-Based Shape Priors	Graph-Based Learning
Noise Robustness	Can reduce small variations because PCA captures dominant shape patterns and ignores some minor variations.	Can learn robust local features, but noisy graph connections or noisy geometric features may affect performance.
Training Data Requirement	Requires a representative set of aligned training shapes.	Usually requires more training data and carefully annotated correspondences.
Implementation	Relatively simple and easy to implement.	More complex because graph construction and graph neural network training are required.
Computational Cost	Generally low and computationally efficient after PCA model construction.	Generally higher because graph operations and neural network training/inference are computationally expensive.
Accuracy	Good for regular and similar shapes with limited variation.	Potentially higher for complex and highly variable 3D face shapes.
Suitability for Morphing	Suitable for simple or controlled morphing applications.	Better suited for realistic morphing involving complex facial expressions and non-rigid deformation.

2. Ability to Handle Non-Rigid Deformations

PCA-Based Shape Priors

PCA learns the major variations present in the training face meshes. The shape can be represented as:

$$x = \bar{x} + Pb$$

Where:

- x = reconstructed shape
- $\bar{x}$ = mean face shape
- P = principal components

- b = shape parameters

PCA can effectively represent common variations such as changes in face width, length, and other regular shape differences.

However, if a face undergoes a complex deformation such as a large change in facial expression, the deformation may not be represented accurately if it was not present in the training data.

Graph-Based Learning

Graph-based learning represents a face mesh as:

$$G=(V,E)$$

Where:

- V = mesh vertices
- E = connections between neighboring vertices

A graph-based model can learn both local geometric relationships and larger structural relationships. Therefore, it can better handle complex non-rigid deformations such as changes around the eyes, mouth, cheeks, and jaw.

Result: Graph-based learning has an advantage for complex non-rigid facial deformations.

3. Robustness to Varying Mesh Resolution and Noise

PCA-Based Shape Priors

PCA generally requires the shapes to be represented in a compatible form. If two meshes have different numbers of vertices, preprocessing such as:

- Resampling
- Landmark extraction
- Mesh alignment
- Normalization

may be required.

PCA can be relatively robust to small noise because it focuses on the major variations in the training dataset.

Graph-Based Learning

Graph-based methods can work with meshes having different numbers of vertices because each mesh is represented as a graph. The model can learn features from the relationships between neighboring vertices.

However, strong noise can affect the graph structure and extracted geometric features. Suitable preprocessing and robust graph features are therefore required.

Result: Graph-based learning is generally more flexible for varying mesh resolutions, while PCA can be simpler and reasonably robust to small noise after alignment.

4. Ease of Implementation and Computational Cost

PCA-Based Shape Priors

PCA is comparatively easy to implement using standard mathematical and machine-learning libraries.

Advantages:

- Simple algorithm
- Lower computational cost
- Easy to train
- Easy to interpret
- Suitable for smaller datasets

Therefore, PCA is a good choice when computational resources are limited.

Graph-Based Learning

Graph-based learning requires:

- Mesh-to-graph conversion
- Graph feature extraction
- Graph neural network architecture
- Training and validation

- More computational resources

Therefore, implementation is more complex and computationally expensive than PCA.

5. Critical Analysis: Which Approach Provides More Accurate Correspondences?

For realistic 3D human face morphing, graph-based learning is generally the better choice.

Reasons

- 1. Handles non-rigid deformation better**
Human faces can undergo complex changes due to facial expressions. Graph-based learning can model local and global geometric relationships.
- 2. Works with varying mesh structures**
The input meshes may contain different numbers of vertices and different connectivity. Graph representations provide greater flexibility.
- 3. Captures local facial features**
Important regions such as the eyes, nose, mouth, and jaw contain complex geometric structures. Graph-based methods can learn relationships between neighboring vertices in these regions.
- 4. Better generalization to complex shapes**
With sufficient training data, graph-based models can learn correspondence patterns across different face shapes and expressions.
- 5. More suitable for realistic morphing**
Accurate correspondence is essential for smooth transitions between facial meshes. Graph-based learning can provide more detailed point-to-point relationships.

6. Advantages and Limitations

PCA-Based Shape Priors

Advantages:

- Simple and easy to implement
- Low computational cost
- Effective for similar face shapes
- Works well with limited variations

- Can reduce the effect of small noise

Limitations:

- Requires aligned training shapes
- Less flexible for different mesh topologies
- May not capture complex non-rigid deformations
- Performance depends on the training shape variations

Graph-Based Learning

Advantages:

- Handles complex non-rigid deformations
- More flexible with varying mesh resolutions
- Learns local and global geometric relationships
- Better suited for complex facial correspondence
- Can provide accurate correspondences for realistic morphing

Limitations:

- More difficult to implement
- Higher computational cost
- Requires suitable training data
- Sensitive to poor-quality graph construction or noisy geometric features

7. Final Recommendation

Requirement	Preferred Approach
Simple implementation	PCA
Low computational cost	PCA
Small shape variations	PCA
Complex non-rigid deformation	Graph-Based Learning
Different mesh resolutions	Graph-Based Learning
Different mesh topologies	Graph-Based Learning
Realistic facial morphing	Graph-Based Learning
Highly accurate correspondence	Graph-Based Learning